

JAVA LAB FILE



SUBMITTED BY:-

NAME : SHAURA PUNDIR

UNIV. ROLL NO : 2021975

CLASS ROLL NO : 56

SECTION : AI

SUBMITTED TO :-

Mr. Vedit

```
/*
```

Name : Shaurya Pundir

University Roll Number: 2021975

Section:AI

Problem 1: Write a java program to take input as a command line argument. Your name, course, university rollno and semester. Display the information as.

Name:

UniversityRollNo:

Course:

Semester

```
*/
```

```
public class p1main {
```

```
    public static void main(String[] args) {
```

```
        int n=args.length;
```

```
        if(n<4)
```

```
        {
```

```
            System.out.println("The details are not sufficient");
```

```
        }
```

```
        else
```

```
        {
```

```
            System.out.println("Name:"+args[0]);
```

```
            System.out.println("University Roll No :"+args[1]);
```

```
            System.out.println("Course:"+args[2]);
```

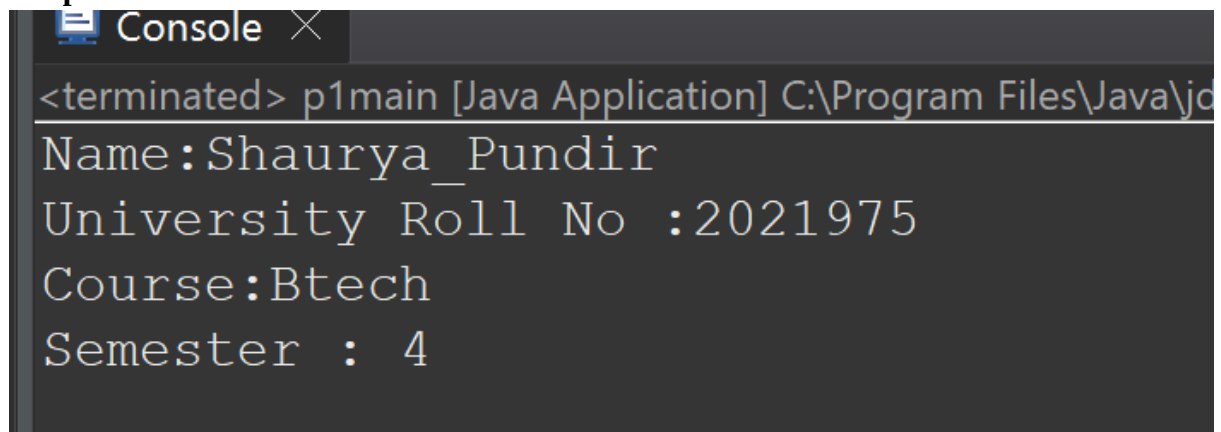
```
            System.out.println("Semester : "+args[3]);
```

```
        }
```

```
    }
```

```
}
```

Output

A screenshot of a Java console window titled "Console" with a close button. The window shows the output of a Java program. The first line is "<terminated> p1main [Java Application] C:\Program Files\Java\j...". The subsequent lines are the program's output: "Name:Shaurya_Pundir", "University Roll No :2021975", "Course:Btech", and "Semester : 4".

```
<terminated> p1main [Java Application] C:\Program Files\Java\j...
Name:Shaurya_Pundir
University Roll No :2021975
Course:Btech
Semester : 4
```

/*

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University Roll Number: 2021975

Section:AI

2) Write a Program to replace all 0's with 1 in a given integer. Given an integer as an input, all the 0's in the number has to be replaced with 1.

For example, consider the following number

Input: 102405

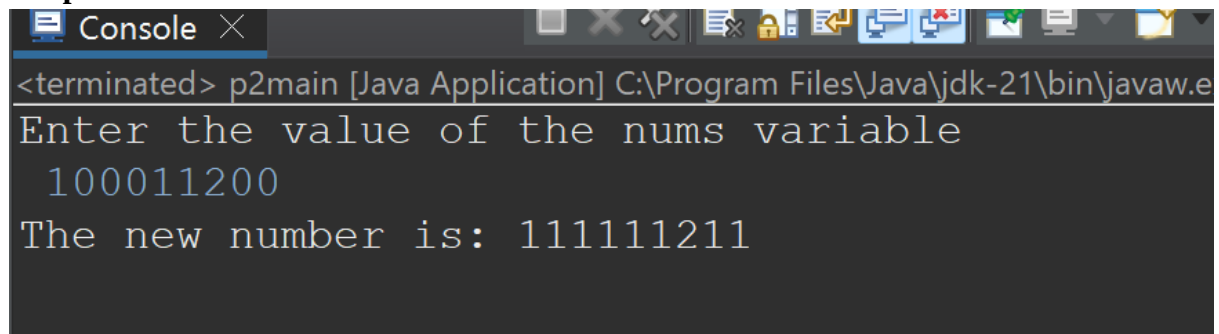
Output: 112415

Input: 56004

Output: 56114*/

```
import java.util.Scanner;
public class p2main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the value of the nums variable");
        int n = sc.nextInt();
        String s = Integer.toString(n);
        s = s.replace('0', '1');
        int m = Integer.parseInt(s);
        System.out.println("The new number is: " + m);
    }
}
```

Output



```
<terminated> p2main [Java Application] C:\Program Files\Java\jdk-21\bin\javaw.e
Enter the value of the nums variable
100011200
The new number is: 111111211
```

/*

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Section:AI

3) Program to find the saddle point coordinates in a given matrix. A saddle point is an element of the matrix, which is the minimum element in its row and the maximum in its column. For example, consider the matrix given below :-Mat[3][3]

1 2 3

4 5 6

7 8 9 Here, 7 is the saddle point because it is the minimum element in its row and maximum element in its column.*/

```
public class SaddlePoint {
    public static void main(String[] args) {
        int[][] matrix = { {1, 2, 3},{4, 5, 6},{7, 8, 9}};
        findSaddlePoint(matrix);
    }
    public static void findSaddlePoint(int[][] matrix) {
        int rows = matrix.length;
        int cols = matrix[0].length;
        for (int i = 0; i < rows; i++) {
            int minRowValue = matrix[i][0];
            int minColIndex = 0;
            for (int j = 1; j < cols; j++) {
                if (matrix[i][j] < minRowValue) {
                    minRowValue = matrix[i][j];
                    minColIndex = j;
                }
            }
            boolean isSaddlePoint = true;
            for (int k = 0; k < rows; k++) {
                if (matrix[k][minColIndex] > minRowValue) {
                    isSaddlePoint = false;
                    break;
                }
            }
            if (isSaddlePoint) {
                System.out.println("Saddle point found at: (" + i + ", " + minColIndex + ")");
                System.out.println("Saddle point value: " + minRowValue);
                return;
            }
        }
        System.out.println("No saddle point found in the matrix.");
    }
}
```

Output

```
<terminated> p3main [Java Application] C:\Program Files\Java\jdk-21\bin\javaw
Saddle point found at: (2, 0)
Saddle point value: 7
```

/*

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Section:AI

4) Program to find all the patterns of 0(1+)0 in the given string. Given a string containing 0's and 1's, find the total number of 0(1+)0 patterns in the string and output it.

0(1+)0 - There should be at least one '1' between the two 0's.

For example, consider the following string.

Input: 01101111010

Output: 3

Explanation:

01101111010 - count = 1

01101111010 - count = 2

01101111010- count = 3 */

import java.util.*;

public class p4main {

public static void main(String[] args) {

Scanner sc = new Scanner(System.in);

String s = sc.next();

int count = 0;

for (int i = 0; i < s.length() - 2; i++) { // Ensure there's room for '0(1+)0'

if (s.charAt(i) == '0') {

for (int j = i + 1; j < s.length() - 1; j++) {

if (s.charAt(j) == '1') {

for (int k = j + 1; k < s.length(); k++) {

if (s.charAt(k) == '0') {

count++;

i = k - 1; // Move i to k-1 to avoid counting overlapping patterns

break;

}

}

break;

}

}

}

}

System.out.println(s + "-count=" + count);

}}

Output

```
001110010
001110010-count=2
```

/*

Name : Shaurya Pundir

University Roll Number: 2021975

Section:AI

5) Write a java program to delete vowels from given string using StringBuffer class. */

```
package Termwork;
```

```
import java.util.*;
```

```
public class p5main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc=new Scanner(System.in);
```

```
        String s=sc.next();
```

```
        StringBuilder sb=new StringBuilder(s);
```

```
        for(int i=0;i<sb.length();i++)
```

```
        {
```

```
            if(sb.charAt(i)=='a' || sb.charAt(i)=='e' || sb.charAt(i)=='i' || sb.charAt(i)=='o'
```

```
||sb.charAt(i)=='u')
```

```
            {
```

```
                sb.deleteCharAt(i);
```

```
                i--;
```

```
            }
```

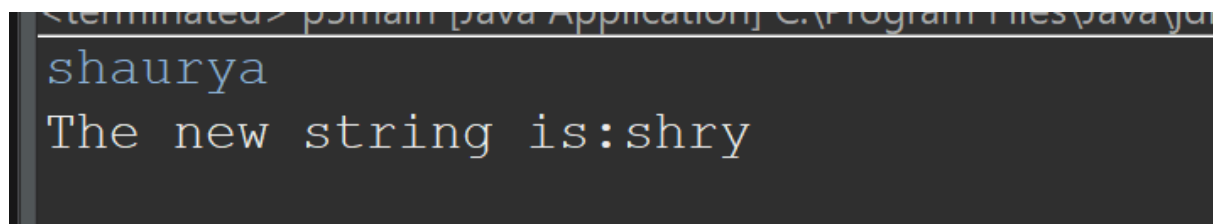
```
        }
```

```
        System.out.println("The new string is:"+sb);
```

```
    }
```

```
}
```

OUTPUT



```
<terminated> p5main [Java Application] C:\Program Files\Java\j...
shaurya
The new string is:shry
```

/*

Name : Shaurya Pundir

University Roll Number: 2021975

Section:AI

6) Write a java program to create a class named 'Bank ' with the following data members:

Name of depositor

Address of depositor

Account Number

Balance in account

Class 'Bank' has a method for each of the following:

1 - Generate a unique account number for each depositor

For first depositor, account number will be 1001, for second depositor it will be 1002 and so on

2 - Display information and balance of depositor

3 - Deposit more amount in balance of any depositor

4 - Withdraw some amount from balance deposited

5 - Change address of depositor

After creating the class, do the following operations

1 - Enter the information (name, address, account number, balance) of the depositors. Number of depositors is to be entered by user.

2 - Print the information of any depositor.

3 - Add some amount to the account of any depositor and then display final information of that depositor

4 - Remove some amount from the account of any depositor and then display final information of that depositor

5 - Change the address of any depositor and then display the final information of that depositor

6 - Randomly repeat these processes for some other bank accounts.

*/

```
package Termwork;
```

```
import java.util.ArrayList;
```

```
import java.util.Scanner;
```

```
class Bankk {
```

```
    private static int nextAccountNumber = 1001;
```

```
    private String name;
```

```
    private String address;
```

```
    private int accountNumber;
```

```
    private double balance;
```

```
    public Bankk(String name, String address, double initialBalance) {
```

```
        this.name = name;
```

```
        this.address = address;
```

```
        this.accountNumber = nextAccountNumber++;
```

```
        this.balance = initialBalance;
```

```
}
```

```
    public void displayInfo() {
```

```
        System.out.println("Account Number: " + accountNumber);
```

```
        System.out.println("Name: " + name);
```

```
        System.out.println("Address: " + address);
```

```
        System.out.println("Balance: " + balance);
```

```
}
```

```

// Method to deposit amount
public void deposit(double amount) {
    if (amount > 0) {
        balance += amount;
        System.out.println("Deposited: " + amount);
    } else {
        System.out.println("Invalid deposit amount");
    }
}

// Method to withdraw amount
public void withdraw(double amount) {
    if (amount > 0 && amount <= balance) {
        balance -= amount;
        System.out.println("Withdrawn: " + amount);
    } else {
        System.out.println("Invalid or insufficient amount for withdrawal");
    }
}

// Method to change address
public void changeAddress(String newAddress) {
    address = newAddress;
    System.out.println("Address updated to: " + newAddress);
}

// Getters for account number and name (for identifying accounts)
public int getAccountNumber() {
    return accountNumber;
}

public String getName() {
    return name;
}

public static void main(String[] args) {
    Scanner scanner = new Scanner(System.in);
    ArrayList<Bankk> bankAccounts = new ArrayList<>();

    // Enter the number of depositors
    System.out.print("Enter the number of depositors: ");
    int numDepositors = scanner.nextInt();
    scanner.nextLine(); // Consume newline

    // Enter information for each depositor
    for (int i = 0; i < numDepositors; i++) {
        System.out.print("Enter name of depositor: ");
        String name = scanner.nextLine();
    }
}

```



```

        System.out.print("Enter address of depositor: ");
        String address = scanner.nextLine();
        System.out.print("Enter initial balance: ");
        double balance = scanner.nextDouble();
        scanner.nextLine(); // Consume newline
        Bankk account = new Bankk(name, address, balance);
        bankAccounts.add(account);
        System.out.println("Account created with account number: " +
account.getAccountNumber());
    }
    boolean continueOperations = true;
    while (continueOperations) {
        System.out.println("\nChoose an operation:");
        System.out.println("1. Display information of a depositor");
        System.out.println("2. Deposit amount");
        System.out.println("3. Withdraw amount");
        System.out.println("4. Change address");
        System.out.println("5. Exit");
        int choice = scanner.nextInt();
        scanner.nextLine(); // Consume newline

        switch (choice) {
            case 1:
                System.out.print("Enter account number: ");
                int accNumberToDisplay = scanner.nextInt();
                scanner.nextLine(); // Consume newline
                for (Bankk account : bankAccounts) {
                    if (account.getAccountNumber() == accNumberToDisplay) {
                        account.displayInfo();
                    }
                }
                break;
            case 2:
                System.out.print("Enter account number: ");
                int accNumberToDeposit = scanner.nextInt();
                System.out.print("Enter amount to deposit: ");
                double depositAmount = scanner.nextDouble();
                scanner.nextLine(); // Consume newline
                for (Bankk account : bankAccounts) {
                    if (account.getAccountNumber() == accNumberToDeposit) {
                        account.deposit(depositAmount);
                        account.displayInfo();
                    }
                }
                break;
            case 3:
                System.out.print("Enter account number: ");
                int accNumberToWithdraw = scanner.nextInt();
                System.out.print("Enter amount to withdraw: ");

```

```

        double withdrawAmount = scanner.nextDouble();
        scanner.nextLine(); // Consume newline
        for (Bankk account : bankAccounts) {
            if (account.getAccountNumber() == accNumberToWithdraw) {
                account.withdraw(withdrawAmount);
                account.displayInfo();
            }
        }
        break;
    case 4:
        System.out.print("Enter account number: ");
        int accNumberToChangeAddress = scanner.nextInt();
        scanner.nextLine(); // Consume newline
        System.out.print("Enter new address: ");
        String newAddress = scanner.nextLine();
        for (Bankk account : bankAccounts) {
            if (account.getAccountNumber() == accNumberToChangeAddress) {
                account.changeAddress(newAddress);
                account.displayInfo(); }
        }
        break;
    case 5:
        continueOperations = false;
        break;
    default:
        System.out.println("Invalid choice, please try again.");
    }
}
scanner.close();
}

```

}Output

```

<terminated> Bankk [Java Application] C:\Program Files\Java\jdk-21\bin\javaw.exe (20-May-2024, 8:10:07 am - 8:10:40 am) [pid: 18328]
Enter the number of depositors: 1
Enter name of depositor: Shaurya
Enter address of depositor: DehraDun
Enter initial balance: 2000
Account created with account number: 1001

Choose an operation:
1. Display information of a depositor
2. Deposit amount
3. Withdraw amount
4. Change address
5. Exit
1
Enter account number: 1001
Account Number: 1001
Name: Shaurya
Address: DehraDun
Balance: 2000.0

Choose an operation:
1. Display information of a depositor
2. Deposit amount
3. Withdraw amount
4. Change address
5. Exit
5

```

/*

Name : Shaurya Pundir

University Roll Number: 2021975

Section:AI

7) Write a Java program to create a class called ArrayDemo and overload arrayFunc() function.

void arrayFunc(int [], int) □ To find all pairs of elements in an Array whose sum is equal to a given number :

Array numbers= [4, 6, 5, -10, 8, 5, 20], target=10

Output :

Pairs of elements whose sum is 10 are :

4 + 6 = 10

5 + 5 = 10

-10 + 20 = 10

*/

package Termwork;

import java.util.Arrays;

class ArrayDemo {

// Overloaded arrayFunc method to find pairs with a specific sum

void arrayFunc(int[] array, int target) {

System.out.println("Pairs of elements whose sum is " + target + " are:");

for (int i = 0; i < array.length; i++) {

for (int j = i + 1; j < array.length; j++) {

if (array[i] + array[j] == target) {

System.out.println(array[i] + " + " + array[j] + " = " + target);

}

}

}

}

void arrayFunc(int[] A, int p, int[] B, int q) {

int[] merged = new int[p + q];

int i = 0, j = 0, k = 0;

// Merge the two arrays

while (i < p && j < q) {

if (A[i] < B[j]) {

merged[k++] = A[i++];

} else {

merged[k++] = B[j++];

}

}

// Copy remaining elements from A

while (i < p) {

merged[k++] = A[i++];

}

```

// Copy remaining elements from B
while (j < q) {
    merged[k++] = B[j++];
}

// Fill A with the first p smallest elements
for (i = 0; i < p; i++) {
    A[i] = merged[i];
}

// Fill B with the remaining elements
for (j = 0; j < q; j++) {
    B[j] = merged[p + j];
}
}

```

```

public static void main(String[] args) {
    ArrayDemo demo = new ArrayDemo();
    int[] numbers = {4, 6, 5, -10, 8, 5, 20};
    int target = 10;

    demo.arrayFunc(numbers, target);
    int[] A = { 1, 5, 6, 7, 8, 10 };
    int[] B = { 2, 4, 9 };
    int p = A.length;
    int q = B.length;

    demo.arrayFunc(A, p, B, q);

    System.out.println("Sorted Arrays:");
    System.out.println("A: " + Arrays.toString(A));
    System.out.println("B: " + Arrays.toString(B));
}
}

```

Output

```

Pairs of elements whose sum is 10 are:
4 + 6 = 10
5 + 5 = 10
-10 + 20 = 10
Sorted Arrays:
A: [1, 2, 4, 5, 6, 7]
B: [8, 9, 10]

```

/*

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Section:AI

8) Write a java program to calculate the area of a rectangle, a square and a circle. Create an abstract class 'Shape' with three abstract methods namely rectangleArea() taking two parameters, squareArea() and circleArea() taking one parameter each.

Now create another class 'Area' containing all the three methods rectangleArea(),squareArea() and circleArea() for printing the area of rectangle, square and circle respectively. Create an object of class Area and call all the three methods.

(Use Runtime Polymorphism)

*/

```
package LAB26;
```

```
abstract class Shape
```

```
{
    int l,b,e;
    float r;
    Shape(int a,int b)
    {
        l=a;
        this.b=b;
    }
    Shape(int e)
    {
        this.e=e;
    }
    Shape(float r)
    {
        this.r=r;
    }
    abstract void area();
}
```

```
class Rectangle extends Shape
```

```
{
    Rectangle(int a1 ,int b1)
    {
        super(a1,b1);
    }
    void area()
    {
        System.out.println("The area of rectangle is: "+l*b);
    }
}
```

```
class Square extends Shape
```

```
{
    Square(int a)
    {
        super(a);
    }
}
```

```

    }
    void area()
    {
        System.out.println("The are of square is :"+e*e);
    }
}
class Circle extends Shape
{
    Circle(float a)
    {
        super(a);
    }
    void area()
    {
        System.out.println("The are of square is :"+2*3.14*r);
    }
}
public class ShapeMain {
    public static void main(String[] args)
    {
        Rectangle r=new Rectangle(10,5);
        Square s=new Square(10);
        Circle c=new Circle((float) 7.0);
        r.area();
        s.area();
        c.area();
    }
}

```

Output

```

The area of rectangle is: 50
The are of square is :100
The are of square is :43.96

```

/*

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University Roll Number: 2021975

Section:AI

9) Write a java program to implement abstract class and abstract method with following details:

Create a abstract Base Class Temperature

Data members:

double temp;

Method members:

void setTempData(double)

abstract void changeTemp()

Sub Class Fahrenheit (subclass of Temperature)

Data members:

double ctemp;

method member:

Override abstract method changeTemp() to convert Fahrenheit temperature into degree Celsius by using formula $C = 5/9 * (F - 32)$ and display converted temperature

Sub Class Celsius (subclass of Temperature)

Data member:

double ftemp;

Method member:

Override abstract method changeTemp() to convert degree Celsius into Fahrenheit temperature by using formula $F = 9/5 * c + 32$ and display converted temperature

*/

package LAB26;

import java.util.Scanner;

```
abstract class Temperature {  
    double temp;
```

```
    void settemp(double temp1) {  
        temp = temp1;  
    }
```

```
    abstract void changetemp();  
}
```

```
class Faren_to_Celci extends Temperature {  
    Faren_to_Celci(double a) {  
        this.settemp(a);  
    }
```

```
    void changetemp() {  
        temp = (5.0 / 9) * (temp - 32);  
        this.settemp(temp);  
        System.out.println("Fahrenheit to Celsius: " + temp);  
    }
```

```
}
```

```

class Celci_to_Faren extends Temperature {
    Celci_to_Faren(double a) {
        this.settemp(a);
    }

    void changetemp() {
        temp = (9.0 / 5) * temp + 32;
        this.settemp(temp);
        System.out.println("Celsius to Fahrenheit: " + temp);
    }
}

public class TemperatureMain {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Convert Fahrenheit to Celsius
        System.out.print("Enter temperature in Fahrenheit: ");
        double fahrenheit = sc.nextDouble();
        Faren_to_Celci farenToCelci = new Faren_to_Celci(fahrenheit);
        farenToCelci.changetemp();

        // Convert Celsius to Fahrenheit
        System.out.print("Enter temperature in Celsius: ");
        double celsius = sc.nextDouble();
        Celci_to_Faren celciToFaren = new Celci_to_Faren(celsius);
        celciToFaren.changetemp();

        sc.close();
    }
}

```

Output

```

Enter temperature in Fahrenheit: 1003
Fahrenheit to Celsius: 539.44444444444445
Enter temperature in Celsius: 43
Celsius to Fahrenheit: 109.4

```



```
/*
```

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Section:AI

10) Write a java program to create an interface that consists of a method to display volume () as an abstract method and redefine this method in the derived classes to suit their requirements.

Create classes called Cone, Hemisphere and Cylinder that implements the interface. Using these three classes, design a program that will accept dimensions of a cone, cylinder and hemisphere interactively and display the volumes.

```
*/
```

```
import java.util.Scanner;
```

```
// Define the interface
```

```
interface Shape {  
    void displayVolume();  
}
```

```
// Class for Cone
```

```
class Cone implements Shape {  
    private double radius;  
    private double height;
```

```
    Cone(double radius, double height) {  
        this.radius = radius;  
        this.height = height;  
    }
```

```
    @Override
```

```
    public void displayVolume() {  
        double volume = (Math.PI * radius * radius * height) / 3;  
        System.out.println("Volume of the Cone: " + volume);  
    }  
}
```

```
// Class for Hemisphere
```

```
class Hemisphere implements Shape {  
    private double radius;
```

```
    Hemisphere(double radius) {  
        this.radius = radius;  
    }
```

```
    @Override
```

```
    public void displayVolume() {  
        double volume = (2 * Math.PI * Math.pow(radius, 3)) / 3;  
        System.out.println("Volume of the Hemisphere: " + volume);  
    }  
}
```

```

class Cylinder implements Shape {
    private double radius;
    private double height;
    Cylinder(double radius, double height) {
        this.radius = radius;
        this.height = height;
    }
    public void displayVolume() {
        double volume = Math.PI * radius * radius * height;
        System.out.println("Volume of the Cylinder: " + volume);
    }
}

public class VolumeCalculator {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter radius of the cone: ");
        double coneRadius = scanner.nextDouble();
        System.out.print("Enter height of the cone: ");
        double coneHeight = scanner.nextDouble();
        Cone cone = new Cone(coneRadius, coneHeight);
        System.out.print("Enter radius of the hemisphere: ");
        double hemisphereRadius = scanner.nextDouble();
        Hemisphere hemisphere = new Hemisphere(hemisphereRadius);
        System.out.print("Enter radius of the cylinder: ");
        double cylinderRadius = scanner.nextDouble();
        System.out.print("Enter height of the cylinder: ");
        double cylinderHeight = scanner.nextDouble();
        Cylinder cylinder = new Cylinder(cylinderRadius, cylinderHeight);
        System.out.println();
        cone.displayVolume();
        hemisphere.displayVolume();
        cylinder.displayVolume();
        scanner.close();
    }
}

```

Output

```

Enter radius of the cone: 23.4
Enter height of the cone: 2.43
Enter radius of the hemisphere: 21.2
Enter radius of the cylinder: 2
Enter height of the cylinder: 23

Volume of the Cone: 1393.3704834536977
Volume of the Hemisphere: 19955.66461817547
Volume of the Cylinder: 289.02652413026095

```

/*

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Section:AI

11) Write a java program to accept and print the employee details during runtime. The details will include employee id, name, dept_Id. The program should raise an exception if user inputs incomplete or incorrect data. The entered value should meet the following conditions:

(i) First Letter of employee name should be in capital letter.

(ii) Employee id should be between 2001 and 5001

(iii) Department id should be an integer between 1 and 5.

If the above conditions are not met then the application should raise specific exception else should complete normal execution.*/

```
package LAB;
```

```
import java.util.*;
```

```
class Emp
```

```
{
```

```
    String name;
```

```
    int eid,deptid;
```

```
    void display(String name,int eid,int deptid)
```

```
    {
```

```
        this.name=name;
```

```
        this.eid=eid;
```

```
        this.deptid=deptid;
```

```
        try
```

```
        {
```

```
            if(!(name.charAt(0)>=65 && name.charAt(0)<=90))
```

```
            {
```

```
                throw new ArithmeticException("The first letter is not Capital");
```

```
            }
```

```
        }
```

```
        catch(Exception e)
```

```
        {
```

```
            System.out.println(e);
```

```
        }
```

```
        try
```

```
        {
```

```
            if(!(eid>=2001 && eid<=5001))
```

```
            {
```

```
                throw new Exception("The employee id id not in range of 2001 - 5001.");
```

```
            }
```

```
        }
```

```
        catch(Exception e)
```

```
        {
```

```
            System.out.println(e);
```

```
        }
```

```
        try
```

```
        {
```

```

        if(!(deptid>=1 && deptid<=5))
        {
            throw new Exception("The dept id is not in the rage of 1 - 5.");
        }
    }
    catch(Exception e)
    {
        System.out.println(e);
    }
    System.out.println("Name :"+name +"\nEmp id: "+eid+"\nDept ID: "+deptid);

}
}
public class EmployeeMain {

    public static void main(String[] args)
    {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the Employee name ,Employee id and Department ID:");
        String name=sc.next();
        int eid=sc.nextInt();
        int deptid=sc.nextInt();
        Emp e=new Emp();
        e.display(name,eid,deptid);
    }
}

```

Output

```

Enter the Employee name ,Employee id and Department ID:
Shaurya 2022
2
Name :Shaurya
Emp id: 2022
Dept ID: 2

```

/*

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Section:AI

12) Create a class MyCalculator which consists of a single method power(int, int).

This method takes two integers, n and p, as parameters and finds np .

If either n or p is negative, then the method must throw an exception which says "n and p should be non-negative".

Input Format

Each line of the input contains two integers, n and p .

Output Format

Each line of the output contains the result ,if neither of n and p is negative.

Otherwise the output contains "n and p should be non-negative".

*/

package LAB;

import java.util.*;

class MyCalculate

```
{
    int n,exp;
    int power(int n,int exp) throws Exception
    {
        if(n<0 || exp<0)
        {
            throw new Exception("The number n & e shouldbe non negative");
        }
        else
        {
            this.n=n;
            this.exp=exp;
            double num=Math.pow(n,exp);
            return (int)num;
        }
    }
}
```

public class MyCalculateMain {

```
    public static void main(String[] args) throws Exception
    {
        MyCalculate obj=new MyCalculate();
        Scanner sc=new Scanner(System.in);
        System.out.print("Enter the value of number and the exponent:");

        try
        {
            int n=sc.nextInt();
            int exp=sc.nextInt();
            int a=obj.power(n,exp);
```

```
        System.out.println(n+"^"+exp+"="+a);
    }

    catch(Exception e)
    {
        System.out.println(e);
        e.printStackTrace();
    }
    finally
    {
        System.out.println("Program Terminated");
    }
}
```

Output

```
Enter the value of number and the exponent:4 2
4^2=16
Program Terminated
```

/*

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Section:AI

13) Write a java file handling program to count and display the number of palindrome present in a text file "myfile.txt".*/

```
package LAB;
```

```
import java.io.File;
```

```
import java.io.FileReader;
```

```
import java.io.FileWriter;
```

```
import java.io.BufferedReader;
```

```
import java.io.BufferedWriter;
```

```
import java.io.IOException;
```

```
public class PallindromeMain {
```

```
    public static boolean isPalindrome(String word) {  
        StringBuilder reversed = new StringBuilder(word).reverse();  
        return word.equalsIgnoreCase(reversed.toString());  
    }
```

```
    public static void main(String[] args) {
```

```
        File file = new File("pallin.txt");  
        try {  
            if (file.createNewFile()) {  
                System.out.println("File is Created");  
            } else {  
                System.out.println("There was a problem while creating the file");  
            }  
        } catch (IOException e) {  
            System.out.println("Exception caught in creating the file ");  
            e.printStackTrace();  
        }
```

```
        try {  
            FileWriter writer = new FileWriter(file);  
            writer.write("Hello madam how are you");  
            writer.close();  
        } catch (IOException e) {  
            System.out.println("Error found in writing the file");  
            e.printStackTrace();  
        }
```

```
        try {  
            FileReader fileReader = new FileReader(file);  
            BufferedReader bufferedReader = new BufferedReader(fileReader);  
            String line;
```

```

int palindromeCount = 0;
while ((line = bufferedReader.readLine()) != null)
{
    String[] words = line.split("\\s+");
    for (String word : words) {
        if (isPalindrome(word)) {
            palindromeCount++;
        }
    }
}
bufferedReader.close();
System.out.println("Number of palindrome words: " + palindromeCount);

} catch (IOException e) {
    System.out.println("Error found in reading the file");
    e.printStackTrace();
}
}
}

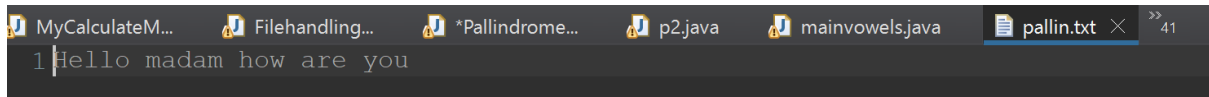
```

Output

```

File is Created
Number of palindrome words: 1

```



The screenshot shows an IDE window with several tabs open: "MyCalculateM...", "Filehandling...", "*Pallindrome...", "p2.java", "mainvowels.java", and "pallin.txt". The "pallin.txt" tab is active, displaying the text "Hello madam how are you" on the first line.

/*

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Section:AI

14) Write a program MultiThreads that creates two threads-one thread with the name CSthread and the other thread named ITthread. Each thread should display its respective name and execute after a gap of 500 milliseconds. Each thread should also display a number indicating the number of times it got a chance to execute.*/

package Termwork;

class THREAD extends Thread {

private String s;

private int count;

THREAD(String s) {

 this.s = s;

 this.count = 0;

}

public void run() {

 try {

 while (count < 5) { // Run 5 times

 sleep(500);

 System.out.println(s + " thread: " + (++count));

 }

 } catch (InterruptedException e) {

 System.out.println("Interrupted Exception caught: " + e);

 }

}

}

public class MultithreadingMAIN {

 public static void main(String[] args) {

 THREAD CS = new THREAD("CS");

 CS.start();

 THREAD IT = new THREAD("IT");

 IT.start();

 }

}

Output

```
CS thread: 1
IT thread: 1
IT thread: 2
CS thread: 2
CS thread: 3
IT thread: 3
IT thread: 4
CS thread: 4
CS thread: 5
IT thread: 5
```

/*

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Section:AI

15) Write a java program for to solve producer consumer problem in which a producer produce a value and consumer consume the value before producer generate the next value.*/

package Termwork;

class SharedBuffer {

private int data;

private boolean hasData = false;

// Producer method

public synchronized void produce(int value) throws InterruptedException {

while (hasData) {

wait(); // Wait until the data is consumed

}

data = value;

hasData = true;

System.out.println("Produced: " + data);

notify(); // Notify the consumer that new data is available

}

// Consumer method

public synchronized void consume() throws InterruptedException {

while (!hasData) {

wait(); // Wait until new data is produced

}

System.out.println("Consumed: " + data);

hasData = false;

notify(); // Notify the producer that the data has been consumed

}

}

class Producer extends Thread {

private SharedBuffer buffer;

public Producer(SharedBuffer buffer) {

this.buffer = buffer;

}

@Override

public void run() {

for (int i = 1; i <= 10; i++) {

try {

buffer.produce(i);

Thread.sleep(100); // Simulate time taken to produce a value

} catch (InterruptedException e) {

e.printStackTrace();

}

}

}

}

```

class Consumer extends Thread {
    private SharedBuffer buffer;

    public Consumer(SharedBuffer buffer) {
        this.buffer = buffer;
    }
    @Override
    public void run() {
        for (int i = 1; i <= 10; i++) {
            try {
                buffer.consume();
                Thread.sleep(150); // Simulate time taken to consume a value
            } catch (InterruptedException e) {
                e.printStackTrace();
            }
        }
    }
}

public class ProducerConsumerMain {
    public static void main(String[] args) {
        SharedBuffer buffer = new SharedBuffer();
        Producer producer = new Producer(buffer);
        Consumer consumer = new Consumer(buffer);
        producer.start();
        consumer.start();
        try {
            producer.join();
            consumer.join();
        } catch (InterruptedException e) {
            e.printStackTrace();
        }
    }
}

```

Output

```

Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5
Produced: 6
Consumed: 6
Produced: 7
Consumed: 7
Produced: 8
Consumed: 8
Produced: 9
Consumed: 9
Produced: 10
Consumed: 10

```

<terminated> ProducerConsumerMain [Java Application] C:\Program Files\Java\jdk-21\bin\javaw.exe (20-May-2024, 10:00:00 AM)

/*

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Section:AI

16) Write a method removeEvenLength that takes an ArrayList of Strings as a parameter and that removes all of the strings of even length from the list.

(Use ArrayList)*/

```
package Termwork;
import java.util.ArrayList;
import java.util.Iterator;
public class RemoveEvenLength {
    public static void removeEvenLength(ArrayList<String> list) {
        Iterator<String> iterator = list.iterator();
        while (iterator.hasNext()) {
            String str = iterator.next();
            if (str.length() % 2 == 0) {
                iterator.remove();
            }
        }
    }
    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>();
        list.add("apple");
        list.add("banana");
        list.add("cherry");
        list.add("date");
        list.add("fig");
        list.add("grape");
        System.out.println("Original list: " + list);
        removeEvenLength(list);
        System.out.println("List after removing even length strings: " + list);
    }
}
```

Output

```
Original list: [apple, banana, cherry, date, fig, grape]
List after removing even length strings: [apple, fig, grape]
```

/*

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Section:AI

17) Write a method swapPairs that switches the order of values in an ArrayList of Strings in a pairwise fashion. Your method should switch the order of the first two values, then switch the order of the next two, switch the order of the next two, and so on.*/

```
import java.util.ArrayList;
```

```
public class SwapPairs {  
    public static void swapPairs(ArrayList<String> list) {  
        for (int i = 0; i < list.size() - 1; i += 2) {  
            String temp = list.get(i);  
            list.set(i, list.get(i + 1));  
            list.set(i + 1, temp);  
        }  
    }  
    public static void main(String[] args) {  
        ArrayList<String> list1 = new ArrayList<>();  
        list1.add("four");  
        list1.add("score");  
        list1.add("and");  
        list1.add("seven");  
        list1.add("years");  
        list1.add("ago");  
        System.out.println("Original list: " + list1);  
        swapPairs(list1);  
        System.out.println("List after swapping pairs: " + list1);  
        ArrayList<String> list2 = new ArrayList<>();  
        list2.add("to");  
        list2.add("be");  
        list2.add("or");  
        list2.add("not");  
        list2.add("to");  
        list2.add("be");  
        list2.add("hamlet");  
        System.out.println("Original list: " + list2);  
        swapPairs(list2);  
        System.out.println("List after swapping pairs: " + list2);  
    }  
}
```

Output

```
Original list: [four, score, and, seven, years, ago]  
List after swapping pairs: [score, four, seven, and, ago, years]  
Original list: [to, be, or, not, to, be, hamlet]  
List after swapping pairs: [be, to, not, or, be, to, hamlet]
```

/*

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Section:AI

18) Write a method called alternate that accepts two Lists of integers as its parameters and returns a new List containing alternating elements from the two lists, in the following order:

- First element from first list
- First element from second list
- Second element from first list
- Second element from second list
- Third element from first list
- Third element from second list
- ...

If the lists do not contain the same number of elements, the remaining elements from the longer list should be placed consecutively at the end. For example, for a first list of (1, 2, 3, 4, 5) and a second list of (6, 7, 8, 9, 10, 11, 12), a call of alternate(list1, list2) should return a list containing (1, 6, 2, 7, 3, 8, 4, 9, 5, 10, 11, 12). Do not modify the parameter lists passed in.*/

```
import java.util.ArrayList;
```

```
import java.util.List;
```

```
public class AlternateElements {
```

```
    public static List<Integer> alternate(List<Integer> list1, List<Integer> list2) {
```

```
        List<Integer> result = new ArrayList<>();
```

```
        int size1 = list1.size();
```

```
        int size2 = list2.size();
```

```
        int maxSize = Math.max(size1, size2);
```

```
        for (int i = 0; i < maxSize; i++) {
```

```
            if (i < size1)
```

```
                result.add(list1.get(i));
```

```
            if (i < size2)
```

```
                result.add(list2.get(i));
```

```
        }
```

```
        return result;
```

```
    }
```

```
    public static void main(String[] args) {
```

```
        List<Integer> list1 = new ArrayList<>();
```

```
        list1.add(1);
```

```
        list1.add(2);
```

```
        list1.add(3);
```

```
        list1.add(4);
```

```
        list1.add(5);
```

```
        List<Integer> list2 = new ArrayList<>();
```

```
        list2.add(6);
```

```
        list2.add(7);
```

```
        list2.add(8);
```

```
        list2.add(9);
```

```
        list2.add(10);
```

```
        list2.add(11);
```

```
list2.add(12);
```

```
List<Integer> result = alternate(list1, list2);
```

```
System.out.println(result);
```

```
}
```

```
}
```

Output

```
[1, 6, 2, 7, 3, 8, 4, 9, 5, 10, 11, 12]
```

/*

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Section:AI

19) Write a GUI program to develop an application that receives a string in one text field, and count number of vowels in a string and returns it in another text field, when the button named “CountVowel” is clicked. When the button named “Reset” is clicked it will reset the value of textfield one and Textfield two. When the button named “Exit” is clicked it will closed the application.*/

```
import javax.swing.*;
```

```
import java.awt.*;
```

```
import java.awt.event.ActionEvent;
```

```
import java.awt.event.ActionListener;
```

```
public class VowelCounterApp extends JFrame {
```

```
    private JTextField inputField;
```

```
    private JTextField outputField;
```

```
    public VowelCounterApp() {
```

```
        setTitle("Vowel Counter");
```

```
        setSize(300, 150);
```

```
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
```

```
        JPanel panel = new JPanel();
```

```
        panel.setLayout(new GridLayout(3, 2));
```

```
        JLabel inputLabel = new JLabel("Input String:");
```

```
        inputField = new JTextField();
```

```
        JLabel outputLabel = new JLabel("Number of Vowels:");
```

```
        outputField = new JTextField();
```

```
        outputField.setEditable(false);
```

```
        JButton countButton = new JButton("CountVowel");
```

```
        countButton.addActionListener(new ActionListener() {
```

```
            @Override
```

```
            public void actionPerformed(ActionEvent e) {
```

```
                String input = inputField.getText().toLowerCase();
```

```
                int count = countVowels(input);
```

```
                outputField.setText(String.valueOf(count));
```

```
            }
```

```
        });
```

```
        JButton resetButton = new JButton("Reset");
```

```
        resetButton.addActionListener(new ActionListener() {
```

```
            @Override
```

```
            public void actionPerformed(ActionEvent e) {
```

```
                inputField.setText("");
```

```
                outputField.setText("");
```

```
            }
```



```
});
```

```
JButton exitButton = new JButton("Exit");  
exitButton.addActionListener(new ActionListener() {  
    @Override  
    public void actionPerformed(ActionEvent e) {  
        dispose();  
    }  
});
```

```
panel.add(inputLabel);  
panel.add(inputField);  
panel.add(outputLabel);  
panel.add(outputField);  
panel.add(countButton);  
panel.add(resetButton);  
panel.add(exitButton);
```

```
add(panel);  
setVisible(true);  
}
```

```
private int countVowels(String input) {  
    int count = 0;  
    for (char c : input.toCharArray()) {  
        if ("aeiou".indexOf(c) != -1) {  
            count++;  
        }  
    }  
    return count;  
}
```

```
public static void main(String[] args) {  
    new VowelCounterApp();  
}
```

Output

```
Error occurred during initialization of boot layer  
java.lang.module.FindException: Error reading module: C:\Users\ACER\eclipse-workspace\c  
Caused by: java.lang.module.InvalidModuleDescriptorException: guii.class found in top-
```